

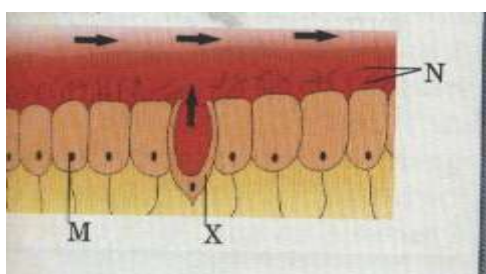
CHIRADZULU DISTRICT MOCK EXAMINATION**2023 MALAWI SCHOOL CERTIFICATE OF EDUCATION EXAMINATION****BIOLOGY****PAPER 1****(100 marks)****Subject number: Mo22/I****Time allowed: 2: hours****Tuesday, 21 March****8:00-10:00 am****Instructions**

1. This paper contains **12** pages.
Please check.
2. Before you begin, **fill** in your **Examination number** at the top of the question paper and on all other sheets.
3. This paper has two sections, A and **B**. Section **A** has ten questions and section **B** has three questions.
4. Answer all the **13** questions in the spaces provided. The maximum number of marks for each answer is indicated against each question.
5. In the table provided on this page, **tick** against the question number you have answered.
6. At the end of the examination, hand in your question paper to the invigilator when time is called to stop writing.

Question number	Tick if answered		
1			
2			
3			
4			
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11			
12			
13			

(2 marks)

2. a. **Figure 1** is a diagram showing a section of a tracheae. Use it to answer the questions that follow.



- (i) Name the parts marked **M** and **N**.

M _____

(1mark)

N _____

(1mark)

- (ii) Mention any **two** functions of the structure in **figure 1** in the human body.

(2 marks)

- b. Explain how the part marked **X** helps in the body defence.

(3 marks)

3. a. (i) Name **one** enzyme produced by the wall of the small intestines.

(1mark)

(ii) What is the end product of digestion if the enzyme mentioned in 3 a (i) above acts on the food substrate?

(1 mark)

b. (i) State **one** function of gastric juice.

(1 mark)

(ii) Describe **one** adaptation of a blood vessel which transports digested food from the small intesines to the liver.

(2 marks)

c. Explain the fate of excess amino acids in the liver.

(3 marks)

4. In chickens, the gene for feather colour has two alleles; the dominant allele, G for grey feathers and the recessive allele, g for white feathers.

(i) What is the genotype of a chicken with grey feathers?

(2 marks)

(ii) What is the genotype of a chicken with homozygous white feathers?

(1 mark)

(iii) What would be the result of a cross between a grey heterozygous with a white chicken. Show your

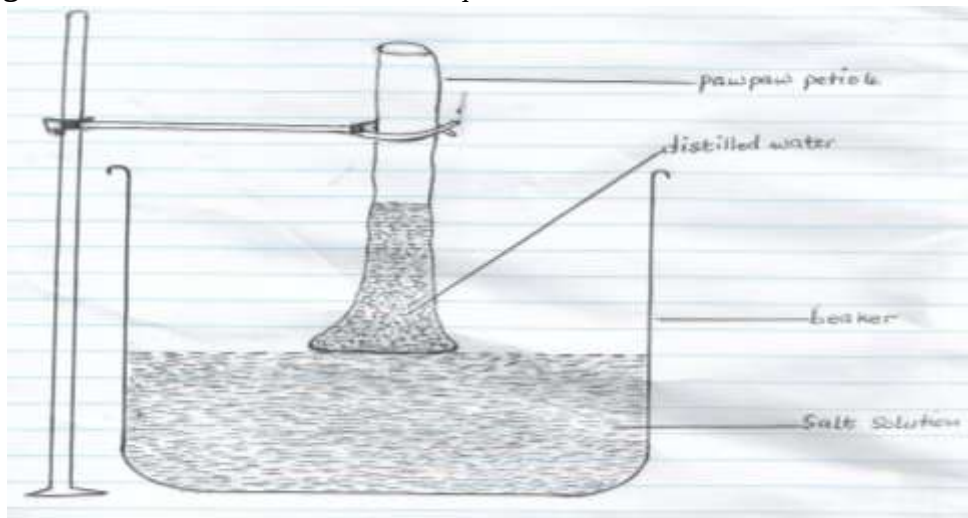
Working.

(4 marks)

(iv) State the phenotypic ratio of F₁ generation for the cross in 4 (iii) above.

(1 mark)

5. Students at a certain school carried out an investigation using pawpaw petiole. They filled distilled water in the cavity of the petiole. They then immersed the base of the petiole in a salt solution contained in the beaker as illustrated in the diagram below. The petiole was supported by a clamp as shown in **figure 2** below. Use it to answer the questions that follow.



- a. What was the role of the petiole base in this investigation?

(1 mark)

- b. Explain why after 20 minutes, the level of water in the petiole decreased.

(3 marks)

- c. Suggest **one** possible control of this investigation.

(1 mark)

6. a. Describe the process of conception.

(2 marks)

- b. Outline any **one** structural difference between a male gamete and a somatic cell of a human being.

(2 marks)

- c. Which of the cells in 6b. above is produced by reduction process?

(1 mark)

7. **Figure 3** is a diagram of a cell organelle, use it to answer the questions that follow.

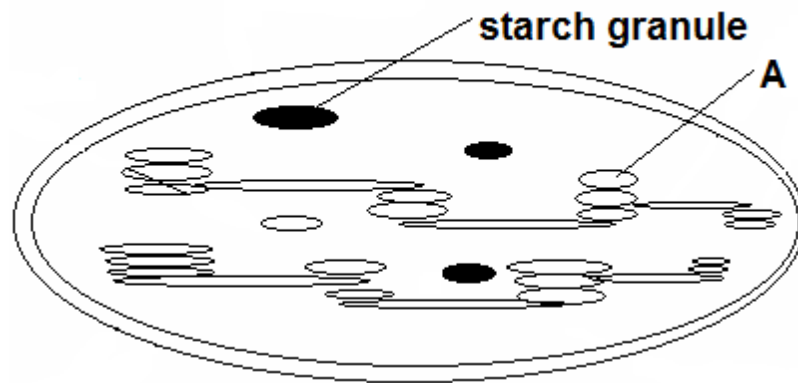


Figure 3: a cell organelle

a. Identify the organelle.

(1 mark)

b. Explain the process that takes place in part labelled A.

(3 marks)

c. Relate this organelle with mitochondrion with respect to their function.

(3 marks)

8. A house maid at Mulanje Boma was running at a speed of 100 m per minute to buy air time at Chitakale Trading Centre. When she crossed Chitakale River, she met a mad dog and her speed immediately rose to 200m per minute.

(i) Mention the hormone that caused the change in speed.

(1 mark)

(ii) What led to the secretion of the hormone in the maid's body?

(1 mark)

(iii) Explain how the hormone led to the change in speed of the maid.

(3 marks)

9. (i). Explain why comparative anatomy can sometimes be described as divergent evolution.

(2 marks)

(ii) Explain how isolation leads to speciation.

(3 marks)

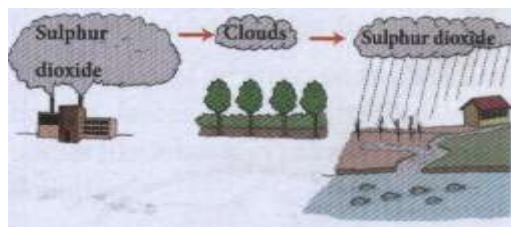
(iii) List any **two** ways how mutations can be beneficial to an organism.

(2 marks)

(iv) In a maize plot, it was observed that all albino maize plants died prematurely. What contributed to this death?

(2 marks)

10. **Figure 4** is a diagram showing a particular cycle. Use it to answer the questions that follow.



(i) Mention **one** process in the diagram that leads to formation of acid rain.

(1 mark)

(ii) Describe any **two** negative effects of acid rains to the environment.

(4 marks)

(iii) State any **one** way of mitigating problems caused by pollutions as shown in **figure 4**.

(1 mark)

(iv) Suggest **one** long lasting solution to the problem of charcoal burning in Malawi.

(1 mark)

SECTION C (30 marks) Essays

11. Explain any **five** ways how xerophytes reduce water loss through transpiration.

[illegible]

(10 marks)

12. Briefly explain five ways of controlling cholera.

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(10 marks)

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13. In an essay form, briefly describe how a dialysis machine works

[illegible]

(10 marks)

End of question paper